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Google Earth Engine based assessment of vegetation, water, and land use dynamics in a diverse mountain–plain transition zone of northeast India

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Abstract

This study assesses multi-temporal land use/land cover (LULC) change and associated vegetation–water dynamics in the Ri-Bhoi district (Meghalaya, NE India), an ecologically sensitive mountain–plain transition zone under increasing development and land-use pressure. Google Earth Engine (GEE) and supervised random forest (RF) classification were applied to Landsat-5 (2010), Landsat-8 (2015), and Sentinel-2 (2020 and 2025) surface reflectance products (cloud threshold $\leq 10\%$). Normalized difference vegetation index and normalized difference water index (NDWI) were derived to evaluate vegetation condition and surface-water variability. The RF classifications achieved overall accuracies of 86–90% with Kappa coefficients of 0.81–0.87 across the four study years, and detailed confusion matrices and publicly accessible GEE scripts are provided in the Supplementary Material (Tables S1–S4; Section S3). Results indicate a persistent decline in forest cover from 1999.61 km² (2010) to 1756.92 km² (2025), corresponding to a net loss of 242.69 km², while agricultural land expanded from 319.49 to 492.92 km² (+ 173.43 km²). Built-up area increased overall (24.54–89.96 km²; + 65.42 km²) but showed inter-annual fluctuations, consistent with class-wise uncertainty discussed in the manuscript. Water bodies varied markedly (26.61 km² in 2010, 44.41 km² in 2015, 22.44 km² in 2020 and 27.14 km² in 2025), reflecting changing hydrological conditions captured by NDWI. Together, the LULC and index trends highlight growing anthropogenic impacts and demonstrate the value of reproducible, cloud-based workflows for regional landscape monitoring and management.

Keywords Google earth engine (GEE), Land use and land cover (LULC) change, Normalized difference vegetation index (NDVI), Normalized difference water index (NDWI), Random forest classification, Landscape dynamics



1 Introduction

Land-use and land-cover (LULC) change in mountain–plain transition zones is often rapid, spatially heterogeneous, and closely linked with ecosystem services, hydrological functioning, and landscape stability. In such environments, forest conversion, agricultural expansion, settlement growth, and modifications in surface-water distribution can significantly influence vegetation condition, surface moisture dynamics, and local environmental quality. Reliable monitoring of these changes over decadal time scales is therefore essential, particularly in regions where complex terrain and mixed land-cover mosaics limit the effectiveness of conventional mapping and change-detection approaches [1–5].

Satellite remote sensing provides an effective basis for assessing multi-temporal LULC dynamics; however, long-term regional analyses are frequently constrained by data volume, preprocessing demands, and the need for consistent workflows across sensors and time periods. Cloud-based geospatial platforms have substantially reduced these barriers by enabling large-scale access to satellite archives, standardized preprocessing, and scalable computation. Among these platforms, Google Earth Engine (GEE) offers an integrated cloud-computing environment with access to multi-sensor image collections and reproducible analysis pipelines, and has been widely adopted for time-series mapping and environmental assessment applications [6, 7]. Previous studies have demonstrated the utility of GEE for regional surface characterization, land surface temperature estimation, fire-related compositing, and surface-water monitoring, highlighting its capacity for consistent, large-area analyses using standardized processing chains [8–10]. Recent GEE-based studies increasingly integrate machine learning and multi-source datasets for hydrometeorological exposure mapping and urban planning applications, underscoring the platform's growing role in reproducible environmental decision-support [11–15].

Vegetation and surface-water dynamics associated with LULC change are commonly examined using spectral indices derived from multispectral imagery. The normalized difference water index (NDWI) enhances open-water features and supports surface-water delineation and monitoring, while vegetation indices based on red and near-infrared reflectance are widely applied to characterize vegetation condition and temporal variability [16–18]. When used alongside LULC classification, such indices provide complementary biophysical context for interpreting land-cover transitions, such as forest-to-agriculture conversion or changes in surface-water extent. Cloud-based computation further facilitates consistent calculation and intercomparison of these indices across years and sensors, improving transparency and repeatability in trend analyses [7, 8].

Accurate multi-temporal LULC mapping in heterogeneous landscapes requires robust classification approaches capable of handling mixed pixels and spectrally similar land-cover classes. Machine-learning techniques, particularly ensemble-based classifiers, are widely applied because they can integrate multispectral information and maintain performance under complex land-cover mosaics [19–23]. Multi-sensor integration and feature-level fusion approaches have also been shown to improve classification reliability, especially where built-up areas, bare land, and agricultural surfaces exhibit overlapping spectral responses [24–26]. Nevertheless, two recurring issues remain in regional multi-temporal LULC mapping: (1) limited reproducibility and cross-sensor comparability due to incomplete reporting of dataset selection, preprocessing thresholds, and classifier settings, and (2) insufficient class-wise uncertainty reporting, which can lead to

over-interpretation of temporal fluctuations for classes prone to confusion in medium-resolution optical imagery (e.g., built-up, bare land, and cropland) [19–23]. These issues are especially relevant in mountain–plain transition zones, where spatial variability in land cover is high and sensor differences across time can bias change estimates if not explicitly addressed [8, 26]. While some studies use expanded feature sets (indices, texture, topography) to improve separability, a conservative, harmonized predictor set can be preferable when the aim is cross-year comparability and reduced multicollinearity.

Against this background, the present study develops and applies a reproducible GEE-based workflow to quantify LULC change and associated biophysical trends in the Ri-Bhoi district of northeast India using multi-temporal satellite observations for 2010, 2015, 2020, and 2025. Ri-Bhoi district represents an ecologically sensitive mountain–plain transition zone between the Khasi Hills and the Brahmaputra plains, characterized by diverse forest–agricultural mosaics, expanding settlements, and prominent surface-water systems, including the Umiam reservoir and associated river networks. The region has experienced agricultural expansion and infrastructure development, and its moderate spatial extent together with comparatively lower persistent cloud cover enables consistent multi-temporal optical analysis using medium-resolution satellite data.

The proposed workflow integrates (1) consistent multi-sensor data access and pre-processing within GEE (including resampling to a common spatial grid to support cross-year comparability), (2) supervised LULC classification using a Random Forest framework with a harmonized predictor set (Blue, Green, Red, and near-infrared bands), and (3) analysis of NDVI and NDWI trends to support interpretation of vegetation and surface-water dynamics [6–8, 27]. To enhance transparency and address common concerns in multi-temporal mapping studies, the methodology explicitly documents satellite dataset selection, preprocessing criteria, and classifier settings; reports validation using confusion matrices and class-wise accuracy metrics (Supplementary Tables S1–S4); and provides publicly accessible GEE scripts for classification and trend analyses (Supplementary Material, Section S3). The specific objectives of the study are to: (1) map and quantify LULC distribution for four benchmark years using a consistent classification workflow; (2) assess the magnitude and direction of major net LULC transitions during 2010–2025; and (3) evaluate NDVI and NDWI trends to complement LULC change interpretation and inform environmental monitoring and land-management planning.

Beyond documenting net class-area changes, this study additionally quantifies gross pixel-to-pixel transitions between 2010 and 2025 (Table 8), allowing identification of dominant conversion pathways (e.g., forest-to-agriculture and forest-to-built-up) that are not captured by net summaries alone. We further derive an annual (2010–2025) NDVI and NDWI time series and apply formal trend tests (Fig. 7, Table 9), complemented by CHIRPS v3 rainfall context (Figs. 6, 8), to strengthen interpretation beyond the four benchmark maps. Together with explicit reporting of sample distribution (Fig. 3), exact sample counts (Table 5), and class-wise uncertainty (Table S10), the workflow provides a transparent and transferable template for ecohydrological and LULC monitoring in mountain–plain transition zones.

2 Study area

The Ri-Bhoi district, covering an area of approximately 2448 km², is located in the northern part of Meghalaya, northeast India, between latitudes 25° 40' N–26° 20' N and longitudes 91° 20' 30" E–92° 17' 00" E. The district forms a transitional zone between the Shillong Plateau (Khasi Hills) to the south and the Brahmaputra plains to the north, making it a representative mountain–plain interface. Ri-Bhoi is bounded by Kamrup district of Assam to the north, Karbi Anglong to the east, East Khasi Hills to the south, and West Khasi Hills to the west. The administrative headquarters is Nongpoh, and the district is divided into three development blocks: Umling, Umsning, and Jirang. The area is well connected by National Highway 40, which links Guwahati and Shillong (Fig. 1) (CGWB, 2013).

The district exhibits a mixed land-use pattern dominated by forests, agriculture, and expanding settlements. Forest cover accounts for approximately 36% of the total area, while agriculture represents the primary livelihood for the predominantly rural population. The net sown area is about 222.6 km², with a total cropped area of approximately 251.7 km², and fallow land occupies nearly 6% of the district (CGWB, 2013). This heterogeneous land-cover composition, combined with ongoing agricultural expansion and infrastructure development, makes Ri-Bhoi particularly suitable for investigating multi-temporal LULC dynamics.

Climatically, Ri-Bhoi experiences a humid subtropical regime with strong monsoonal influence. The district receives an average annual rainfall of approximately 2935 mm, with pronounced seasonal variability. Mean temperatures range from around 10 °C during winter months to approximately 30 °C in summer, with occasional higher extremes recorded at lower elevations (CGWB, 2013). These climatic conditions strongly influence vegetation productivity and surface-water availability, thereby affecting NDVI and NDWI patterns across seasons and years.

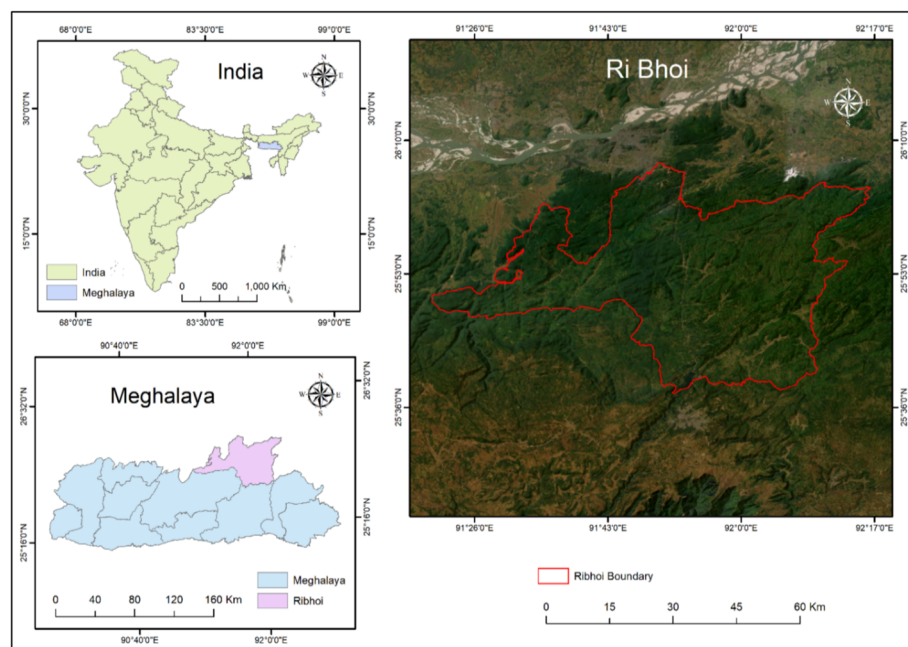


Fig. 1 Location map of the study area (Ri-Bhoi district)

Geomorphologically, the district is characterized by hilly terrain interspersed with intermontane valleys and dissected plateaus. The southern and central parts comprise structurally controlled hills and valleys developed over Precambrian formations of the Shillong Plateau, while the northern sector grades into gentler terrain associated with the Brahmaputra valley. The drainage network is structurally influenced and includes major rivers such as the Umtrew, Umsiang, Umran, and Umiam, along with the prominent Umiam reservoir. These geomorphic and hydrological characteristics contribute to spatial variability in surface-water distribution and vegetation cover, which are critical for interpreting LULC transitions and associated biophysical trends (CGWB, 2013).

Overall, the combination of complex topography, diverse land-cover mosaics, monsoon-driven hydrology, and increasing anthropogenic pressure makes Ri-Bhoi district an appropriate and representative study area for assessing multi-temporal LULC change and NDVI/NDWI dynamics using medium-resolution satellite data within a cloud-based analytical framework.

3 Data set and methodological approach

3.1 Data collection and platform

Google Earth Engine (GEE) was used as the primary platform for satellite data access, preprocessing, and analysis. GEE provides cloud-based access to multi-sensor Earth observation datasets and enables scalable, reproducible geospatial processing [6, 7, 28, 29]. In this study, surface reflectance imagery from the Landsat and Sentinel missions was accessed through GEE archives maintained by the United States Geological Survey (USGS) and the European Space Agency (ESA).

The satellite datasets, corresponding GEE product IDs, target years, and cloud-cover thresholds applied in the analysis are summarized in Table 1. To ensure data quality and inter-annual comparability, only scenes with cloud cover $\leq 10\%$ were selected for all study years.

To contextualize NDWI variability and explore hydro-meteorological influence, benchmark-year precipitation was derived from CHIRPS v3 rainfall products available in GEE and summarized as spatial rainfall surfaces and an inter-annual trendline (Figs. 6, 8). Similar GEE-based studies have used gridded precipitation products and remote-sensing indicators to support regional rainfall exposure and water-related planning analyses [12–15].

3.2 Methodological framework

The overall methodological workflow adopted in this study is illustrated in Fig. 2. The framework integrates satellite data preprocessing, computation of vegetation and water indices, supervised LULC classification using a Random Forest algorithm, and accuracy

Table 1 Satellite data sources and preprocessing criteria used in the study

Data layer	Platform/product ID	Target year	Cloud threshold (%)
Landsat 5 TM Surface Reflectance (Tier 1)	LANDSAT/LT05/C02/T1_L2	2010	≤ 10
Landsat 8 OLI Surface Reflectance (Tier 1)	LANDSAT/LC08/C02/T1_L2	2015	≤ 10
Sentinel-2 MSI Level-2A (Harmonized)	COPERNICUS/S2_SR_HARMONIZED	2020	≤ 10
Sentinel-2 MSI Level-2A (Harmonized)	COPERNICUS/S2_SR_HARMONIZED	2025	≤ 10

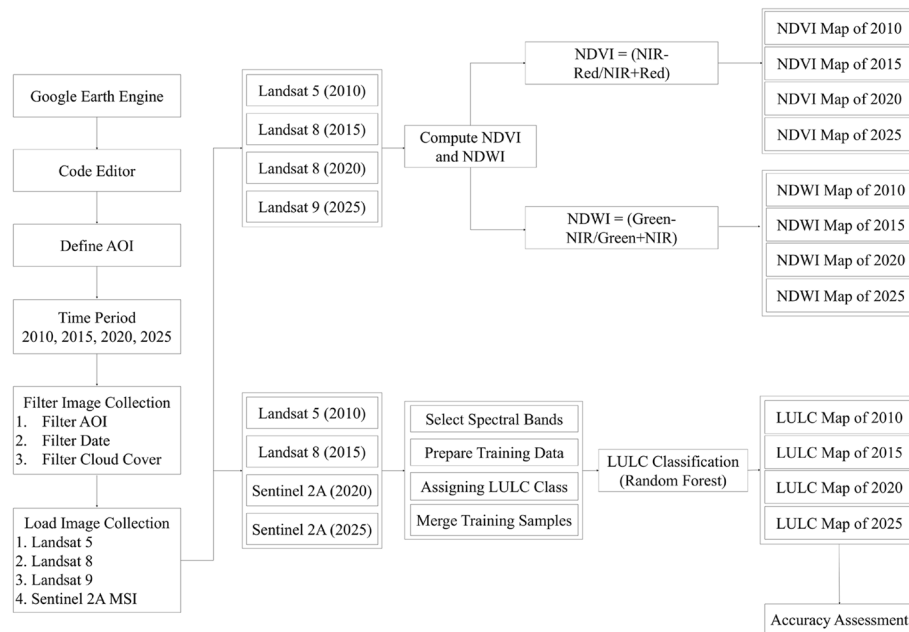


Fig. 2 Methodological workflow for LULC classification and computation of NDVI and NDWI using the Google Earth Engine platform

assessment. All processing steps were implemented within the GEE environment to ensure methodological consistency and reproducibility across years.

3.2.1 Satellite data preprocessing

Orthorectified surface reflectance imagery with minimal cloud contamination was used as the primary input dataset. Cloud and cloud-shadow masking were performed within GEE using quality assessment bands and established masking approaches following García et al. [30], Simonetti et al. [31], and Zurqani et al. [32].

For each target year (2010, 2015, 2020, and 2025), cloud-free annual composites were generated using a median reducer, which calculates the median reflectance value for each pixel across all valid observations within the specified time window. This approach reduces the influence of residual cloud contamination and extreme reflectance values. Landsat-5 TM imagery was used for 2010, Landsat-8 OLI for 2015, and Sentinel-2 MSI surface reflectance data for 2020 and 2025. The spectral characteristics of the datasets used are summarized in Table 2. Final map layouts were prepared using ArcGIS Pro for visualization purposes. All inputs were taken from surface reflectance products in GEE; no additional standardization (e.g., z-score normalization) was applied prior to classification, and RF models were trained independently for each target year. To reduce spatial-resolution mismatch between sensors, Sentinel-2 composites (native 10 m bands) were resampled to 30 m within GEE to match the Landsat grid prior to classification and area estimation. To minimize spectral-response mismatch, the classification was restricted to the four common surface-reflectance bands (Blue, Green, Red, and Near-Infrared) available across Landsat and Sentinel-2.

Table 2 Spectral characteristics of the satellite data used for assessing LULC and normalized difference indices (NDVI/NDWI) in the study area

Data layer	Source	Bands	Wave-length (μm)	Band-width (μm)	Spatial resolution (m)
Landsat 5 TM (Thematic Mapper), Surface Reflectance Tier 1	Google Earth Engine (GEE), data accessed via the U.S. Geological Survey (USGS)	Blue (Band 1)	0.485	0.07	30
		Green (Band 2)	0.56	0.06	30
		Red (Band 3)	0.66	0.06	30
		Near-Infra-Red (Band 4)	0.835	0.12	30
		Short-Wave Infra-Red 1 (Band 5)	1.65	0.2	30
		Thermal Infrared (Band 6)	11.45	2.1	120 (resampled to 30)
		Short-Wave Infra-Red 2 (Band 7)	2.215	0.27	30
Landsat 8 & 9 OLI (Operational Land Imager), Surface Reflectance Tier 1	Google Earth Engine (GEE), data accessed via the U.S. Geological Survey (USGS)	Coastal Aerosol (Band 1)	0.443	0.02	30
		Blue (Band 2)	0.482	0.06	30
		Green (Band 3)	0.561	0.057	30
		Red (Band 4)	0.655	0.038	30
		Near-Infrared (Band 5)	0.865	0.028	30
		Short-Wave Infrared 1 (Band 6)	1.609	0.085	30
		Short-Wave Infrared 2 (Band 7)	2.2	0.186	30
		Panchromatic (Band 8)	0.59	0.18	15
		Cirrus (Band 9)	1.373	0.02	30
		Thermal Infrared 1 (Band 10)	10.895	0.59	100 (resampled to 30)
		Thermal Infrared 2 (Band 11)	12.005	1.01	100 (resampled to 30)
Sentinel-2 MSI Level-2A (Surface Reflectance)	Google Earth Engine (GEE), data accessed via ESA Copernicus and USGS	Band 1—Coastal aerosol	0.443	0.02	60
		Band 2—Blue	0.496	0.066	10
		Band 3—Green	0.56	0.036	10
		Band 4—Red	0.665	0.031	10
		Band 5—Red Edge 1	0.704	0.015	20
		Band 6—Red Edge 2	0.74	0.015	20
		Band 7—Red Edge 3	0.783	0.02	20
		Band 8—NIR	0.842	0.106	10
		Band 8A—Narrow NIR (Red Edge 4)	0.865	0.021	20
		Band 9—Water vapor	0.945	0.02	60
		Band 10—SWIR (Cirrus)	1.375	0.03	60
		Band 11—SWIR 1	1.61	0.091	20
		Band 12—SWIR 2	2.19	0.175	20

3.2.2 Vegetation and water indices

Vegetation and surface-water dynamics were assessed using the normalized difference vegetation index (NDVI) and the normalized difference water index (NDWI). NDVI quantifies vegetation condition using the normalized difference between near-infrared (NIR) and red reflectance [33], while NDWI enhances surface-water features using the normalized difference between green and NIR reflectance [27]. The indices were calculated as:

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)} \quad (1)$$

$$NDWI = \frac{(Green - NIR)}{(Green + NIR)} \quad (2)$$

These indices were computed consistently across all study years within GEE and were used to support interpretation of LULC changes and biophysical trends over time.

Table 3 Image classification details

Sl. no	Class	Description
1	Built-up	Land covered by concrete, including low-, medium-, and high-density road networks; residential, industrial, and commercial buildings; educational institutes; transportation; open-roof concrete structures; other human-made structures; and solid waste landfills
2	Forest	Land characterized by evergreen and relatively sparse forest vegetation
3	Agriculture land	Areas characterized by a high density of croplands, plantations, fallow land, and current shifting cultivation areas
4	Waste land	Areas with or without sparse vegetation that are likely to change or be converted to other uses in the future. This includes land without crops, barren rocky areas, and sandy stretches along rivers
5	Water bodies	Areas covered by water include rivers, reservoirs, ponds, lakes, and streams

Table 4 Rating index of agreement by kappa coefficient

Sl no	Coefficient range	Accuracy rating
1	0	Equivalent to chance
2	0.10–0.20	Slight
3	0.21–0.40	Fair
4	0.41–0.60	Moderate
5	0.61–0.80	Substantial
6	0.81–0.99	Near perfect
7	1	Perfect

3.2.3 LULC classification

Land-use/land-cover classification was performed using cloud-free annual median composites for each study year. The study adopted a five-class LULC scheme—water bodies, built-up areas, forest land, agricultural land, and wasteland—consistent with USGS Level-1 classification standards. A detailed description of the LULC classes is provided in Table 3. The interpretation of Kappa agreement ratings used in this study is summarized in Table 4.

Training samples for each LULC class were generated within GEE using high-resolution Google Earth imagery as reference. Samples were prepared independently for each target year (2010, 2015, 2020, and 2025) using year-appropriate imagery, following the same class definitions and sampling protocol to maintain comparability. Reference points were distributed across the full study area (Fig. 3). Exact class-wise sample counts for each year, together with the training ($\approx 70\%$) and validation ($\approx 30\%$) subsets used for accuracy assessment, are reported in Table 5.

To improve representativeness and reduce potential spatial autocorrelation, reference points were intentionally dispersed across the full study area and across diverse land-cover contexts (Fig. 3), avoiding dense clustering where possible. Because the training–validation partition is performed at the point level, some residual spatial dependence may remain; therefore, we report class-wise precision/recall and F1-scores (Table S10) and interpret small inter-annual fluctuations conservatively.

3.2.4 Random forest classification

Supervised LULC classification was carried out using the random forest (RF) algorithm, which is widely used in remote sensing due to its robustness to overfitting and ability to handle high-dimensional data [34, 35]. RF constructs an ensemble of decision trees using bootstrapped subsets of training data and randomly selected predictor variables, with final class assignment determined through majority voting [36, 37]. In this study,

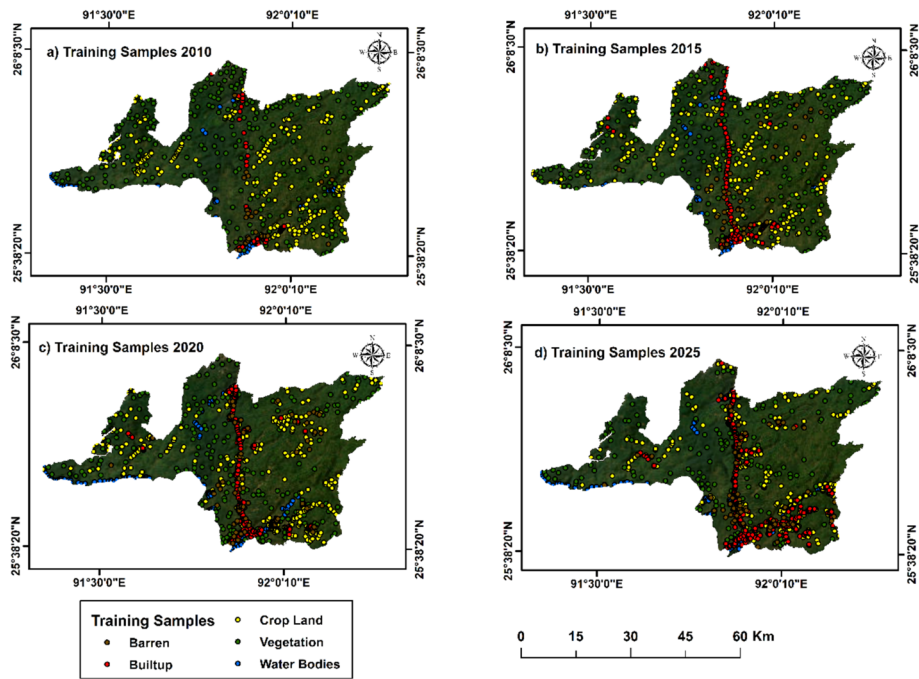


Fig. 3 Spatial distribution of reference/training samples used for supervised RF classification: **a** 2010, **b** 2015, **c** 2020, **d** 2025

Table 5 Exact number of reference points by class and year, including the training–validation split used for RF classification and accuracy assessment

Year	Water bodies	Forest	Agriculture land	Built-up	Waste land	Total
a: Reference samples (all)						
2010	61	249	207	45	23	585
2015	40	242	176	106	38	602
2020	127	215	286	198	78	904
2025	88	160	228	338	67	881
b: Training samples						
2010	43	173	153	35	15	419
2015	24	165	113	72	29	403
2020	89	166	199	139	58	651
2025	57	109	168	236	51	621
c: Validation samples						
2010	18	76	54	10	8	166
2015	16	77	63	34	9	199
2020	38	49	87	59	20	253
2025	31	51	60	102	16	260

RF was implemented within the GEE platform using 500 decision trees (nTrees = 500), while the number of variables per split (mTry) was left at the default setting. Only the four common surface-reflectance spectral bands (Blue, Green, Red, and Near-Infrared) were used as predictor variables; no additional indices (e.g., NDVI/NDWI), texture measures, or topographic variables (elevation/slope) were included. This choice maintains model simplicity and reduces the risk of multicollinearity while still achieving acceptable classification accuracy. The same RF settings and predictor set were used across all years to support inter-annual comparability.

3.2.5 Accuracy assessment

Classification accuracy was evaluated using confusion matrices following Congalton [38]. For each year, the reference-point dataset was split at the point level into training and validation subsets using a nominal 70/30 partition with a fixed random seed (seed=42) to ensure reproducibility; exact class-wise counts for the reference, training, and validation subsets are reported in Table 5. Accuracy metrics including overall accuracy (OA), producer's accuracy (PA), user's accuracy (UA), and the Kappa coefficient (κ) were derived to assess classification performance.

$$\text{Producer's Accuracy (PA)} = \left(\frac{x_{kk}}{x_{+k}} \right) \times 100 \quad (3)$$

$$\text{User's Accuracy (UA)} = \left(\frac{x_{kk}}{x_{k+}} \right) \times 100 \quad (4)$$

$$\text{Overall accuracy (OA)} = \frac{1}{N} \sum_{k=1}^r n_i \quad (5)$$

$$\text{Kappa coefficient (k)} = \frac{N \sum_{k=1}^r x_{kk} - \sum_{k=1}^r (x_{k+} \cdot x_{+k})}{N^2 - \sum_{k=1}^r (x_{k+} \cdot x_{+k})} \quad (6)$$

where N is the sum of pixels; r denotes the number of classes; x_{kk} represent the correct pixels in row "k" and column "k"; x_{k+} and x_{+k} represent the marginal total of row k and column k.

While OA and Kappa values are summarized in the main text (Table 6), full confusion matrices and class-wise accuracy diagnostics for all study years are provided in the Supplementary Material (Tables S1–S4). These diagnostics support cautious interpretation of classes prone to spectral confusion; additional class-wise precision/recall and F1-scores are reported in Supplementary Table S10.

4 Results

4.1 NDVI (2010–2025)

NDVI was used to assess spatio-temporal variation in vegetation condition across the study period. The 2010 NDVI distribution indicated predominantly moderate-to-high vegetation covers across large parts of the district, with comparatively lower NDVI values concentrated around settlement corridors and cultivated patches. In 2015, NDVI patterns remained broadly similar, but localized declines were evident in several transitional zones. By 2020 and 2025, NDVI maps showed increasing spatial heterogeneity, with localized low-to-moderate NDVI zones expanding in areas associated with land conversion and intensifying human activity. Despite these localized declines and fragmentation patterns, the district-wide annual mean NDVI exhibits a significant

Table 6 Accuracy assessment of Random Forest classification for LULC mapping

Year	Overall accuracy (%)	Kappa coefficient
2010	86	0.81
2015	88	0.83
2020	90	0.87
2025	89	0.85

increasing trend over 2010–2025 (Table 9), suggesting that greening in some areas (e.g., agricultural expansion/plantations) offsets localized losses in the mean signal (Fig. 4).

4.2 NDWI (2010–2025)

NDWI was used to characterize surface-water patterns and moisture-related variation across the district. In 2010, NDWI values indicated moderate surface-water presence, primarily associated with major river networks and the Umiam reservoir system. In 2015, NDWI patterns showed increased surface-water signal and/or moisture presence in several zones relative to 2010. In 2020, NDWI values declined markedly, indicating reduced surface-water extent and/or moisture signal during that year (Fig. 5).

By 2025, NDWI showed partial recovery compared to 2020, although the spatial pattern suggests that the hydrological signal remained variable across the district. Overall, the annual mean NDWI exhibits a statistically significant decreasing trend during 2010–2025, and benchmark-year rainfall patterns (Figs. 6, 7, 8) provide climatic context for inter-annual moisture variability.

4.3 NDVI and NDWI trend analysis

To summarize temporal dynamics, annual mean NDVI and NDWI values (2010–2025) were computed and evaluated using both linear regression and Mann–Kendall trend tests (Table 9). NDVI shows a statistically significant positive trend (OLS slope = 0.0070 year⁻¹, $p = 0.0014$; Mann–Kendall $p = 0.013$), whereas NDWI shows a significant negative trend (OLS slope = -0.0048 year⁻¹, $p = 0.0044$; Mann–Kendall $p = 0.027$). These results indicate overall greening alongside a declining surface-wetness signal over the study period, while benchmark-year maps (Figs. 4, 5, 6) illustrate strong spatial heterogeneity and inter-annual variability (Figs. 7, 8).

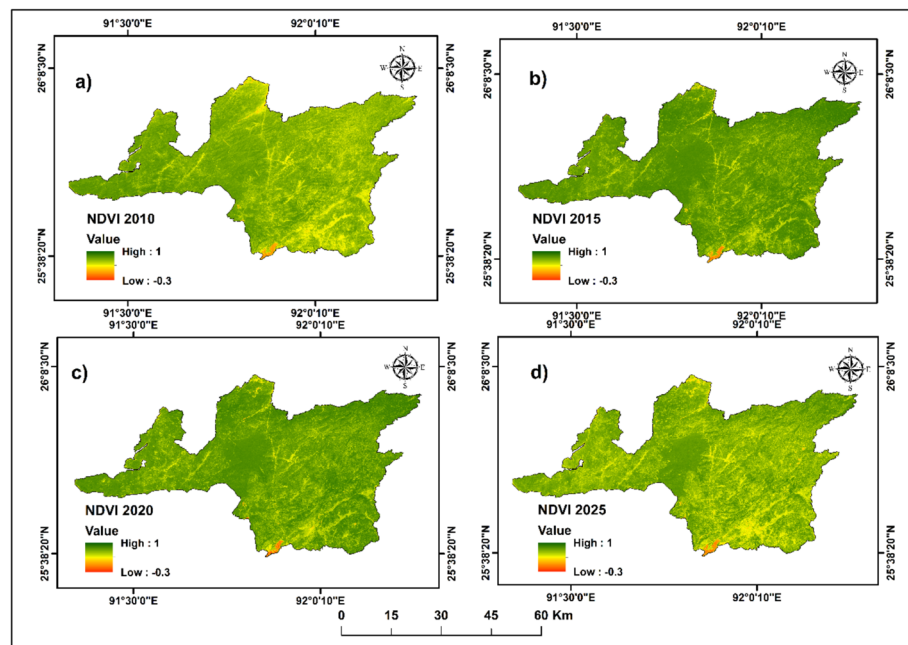


Fig. 4 NDVI maps of Ri-Bhoi district for the years 2010–2025

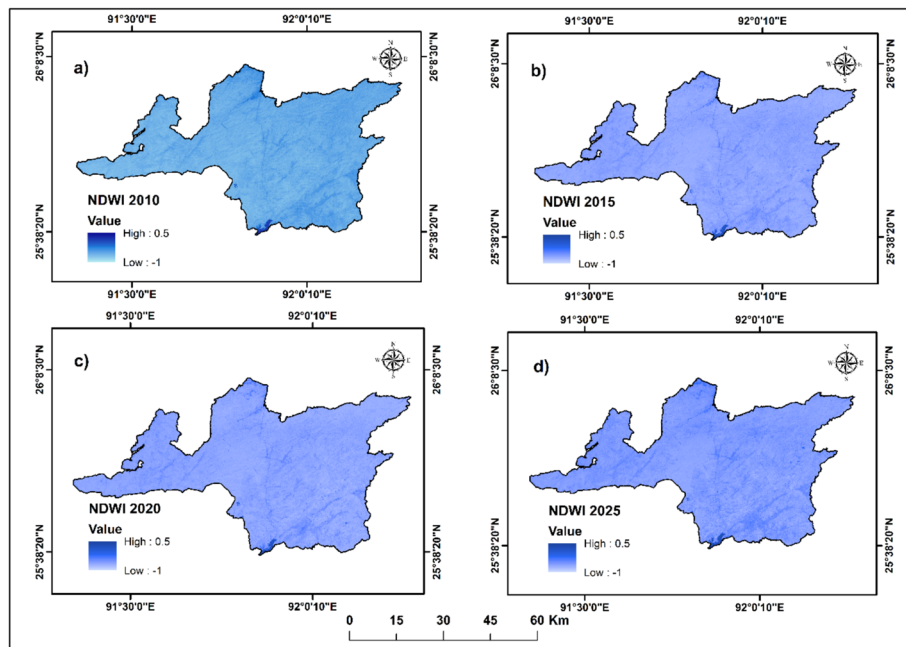


Fig. 5 NDWI maps of Ri-Bhoi district for the years 2010–2025

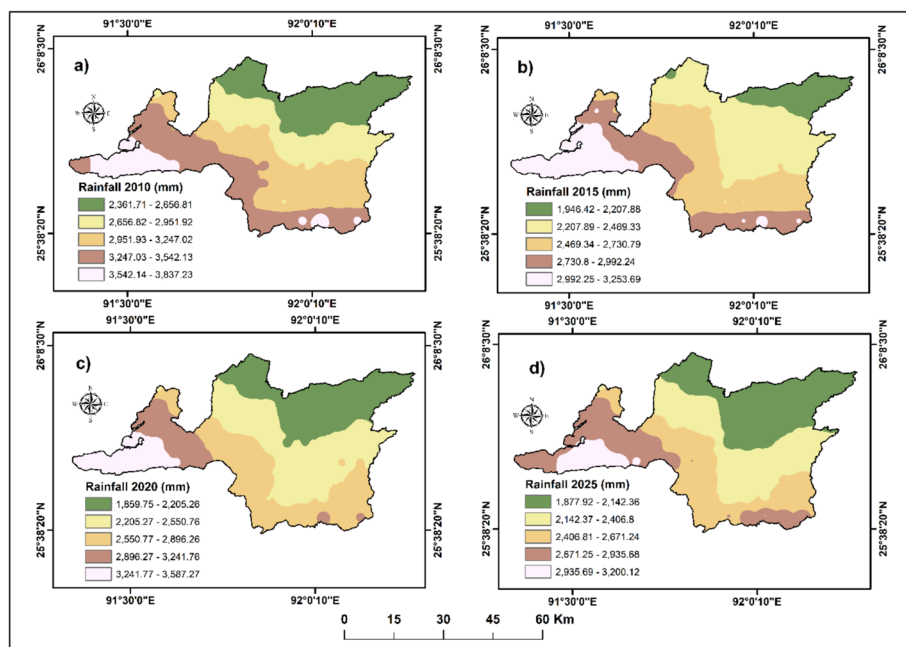


Fig. 6 Spatial distribution of annual rainfall over the study area for benchmark years **a** 2010, **b** 2015, **c** 2020, and **d** 2025. Rainfall surfaces were derived from CHIRPS v3 precipitation (Google Earth Engine) and summarized to annual totals (mm)

4.4 Land use/land cover distribution (2010–2025)

LULC classification was performed for 2010, 2015, 2020, and 2025 using a supervised Random Forest approach implemented in Google Earth Engine. The classified maps indicate that forest land remained the dominant class throughout the period but showed a consistent decline over time. In 2010, forest cover was estimated at 1999.61 km²,

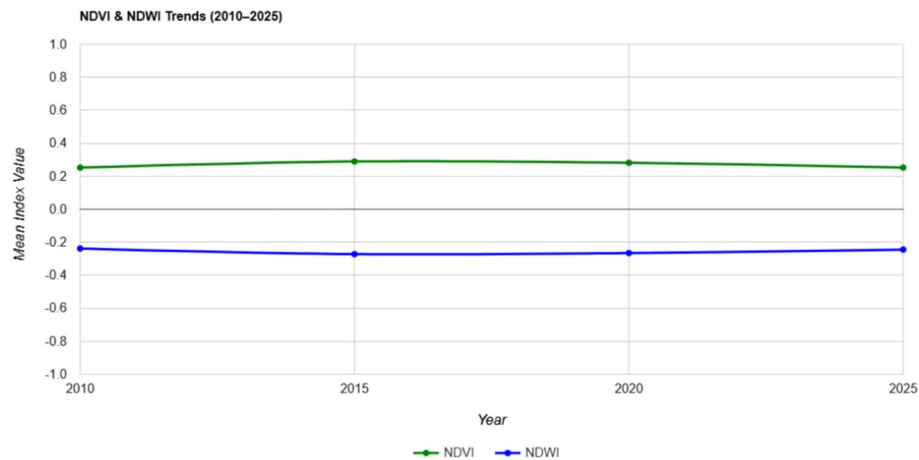


Fig. 7 Annual mean NDVI and NDWI time series (2010–2025) with trend significance (Mann–Kendall/regression)

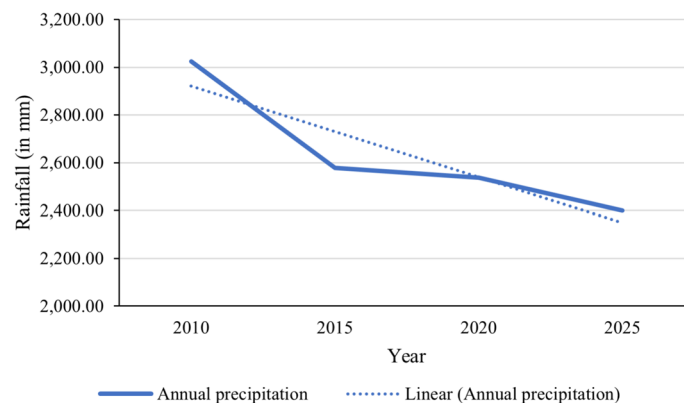


Fig. 8 Trend of annual precipitation over the study area during the benchmark years (2010–2025)

decreasing to 1911.25 km² in 2015 and to 1895.11 km² in 2020, reaching 1756.92 km² in 2025. Agricultural land showed a continuous increase from 319.49 km² (2010) to 323.22 km² (2015), 418.16 km² (2020), and 492.92 km² (2025). Built-up area increased overall from 24.54 km² in 2010 to 89.96 km² in 2025, with fluctuations across intermediate years. Water bodies exhibited temporal variability, with higher estimated area in 2015 (44.41 km²), a decrease in 2020 (22.44 km²), and a partial increase in 2025 (27.14 km²). Wasteland increased from 1.05 km² (2010) to 4.37 km² (2025), although it remained a small fraction of total area.

A temporal trend plot of class areas summarizes these patterns and indicates that the dominant trajectory during 2010–2025 is forest decline and agriculture expansion, while water and built-up show higher inter-annual variability (Figs. 9, 10).

4.5 Land use/land cover change analysis (2010–2025)

Net changes in LULC between 2010 and 2025 indicate strong land-cover transitions (Table 7). Forest land exhibited the largest net decrease (−242.69 km²), while agricultural land showed a major increase (+173.43 km²). Built-up area increased by 65.42 km² over the period, although intermediate-year variability indicates non-linear growth. Water bodies showed comparatively minor net change over 2010–2025 (+0.53 km²) but substantial year-to-year variability. Wasteland increased by 3.32 km². To avoid ambiguity

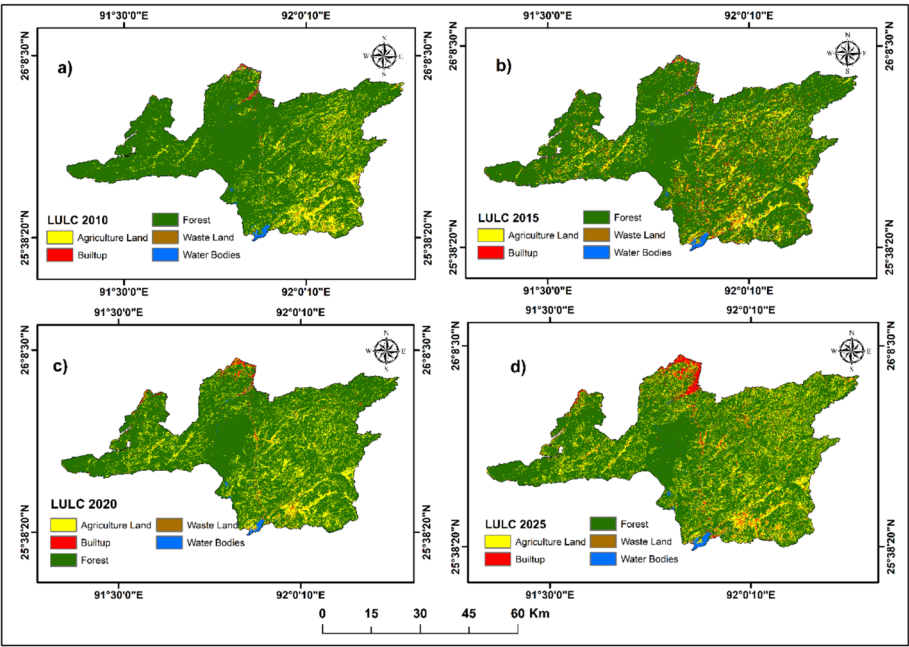


Fig. 9 LULC maps of Ri-Bhoi district for the years 2010–2025

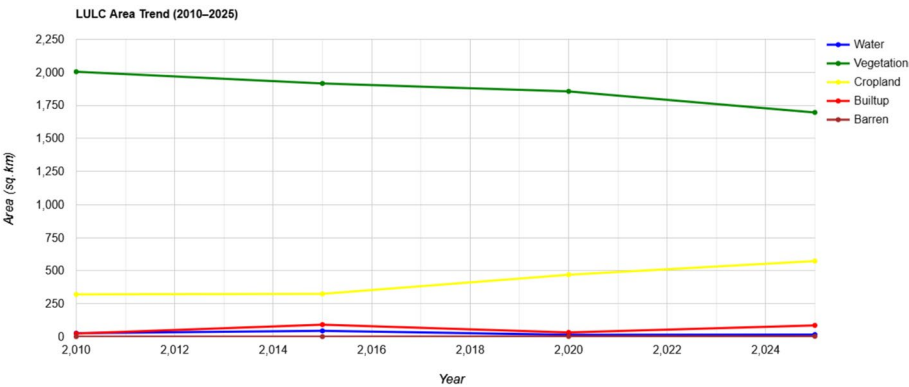


Fig. 10 Temporal trend of LULC class areas across the study period (2010–2025)

Table 7 Net change in LULC from 2010 to 2025 in Ri-Bhoi district						
LULC	2010	2015	2020	2025	Net change (km ²)	Trend (2010–2025)
Water bodies	26.61	44.41	22.44	27.14	0.53	Slight Increase
Forest	1999.61	1911.25	1895.11	1756.92	− 242.69	Decrease
Agriculture land	319.49	323.22	418.16	492.92	173.43	Increase
Built-up	24.54	91.27	32.87	89.96	65.42	Increase (fluctuating)
Waste land	1.05	1.15	2.73	4.37	3.32	Increase

inherent in net-only summaries, a gross pixel-to-pixel transition matrix between 2010 and 2025 (Table 8) quantifies all class-to-class conversions. The largest gross conversion is from forest to agricultural land (315.65 km²), followed by forest to built-up (39.20 km²) and agricultural land to built-up (26.68 km²).

Figure 11 illustrates class-wise area distribution changes for the later period, while Fig. 12 summarizes net changes across the full assessment window. The gross transition

Table 8 Gross post-classification transition matrix between 2010 and 2025 (km²) based on pixel-to-pixel cross-tabulation (rows = 2010 class; columns = 2025 class)

From class	Water bodies	Forest	Agriculture land	Built-up	Waste land	Row total (km ²)
Water bodies	11.150	5.721	5.000	4.697	0.040	26.608
Forest	7.541	1635.505	315.649	39.204	1.713	1999.611
Agriculture land	2.101	115.981	173.068	26.684	1.657	319.491
Built-up	0.797	4.737	8.804	10.161	0.043	24.542
Waste land	0.002	0.130	0.311	0.331	0.278	1.051
Col total (km ²)	21.590	1762.073	502.832	81.077	3.731	2371.304

Areas are computed within the common valid pixel footprint of the 2010 and 2025 classified rasters (masked pixels excluded)

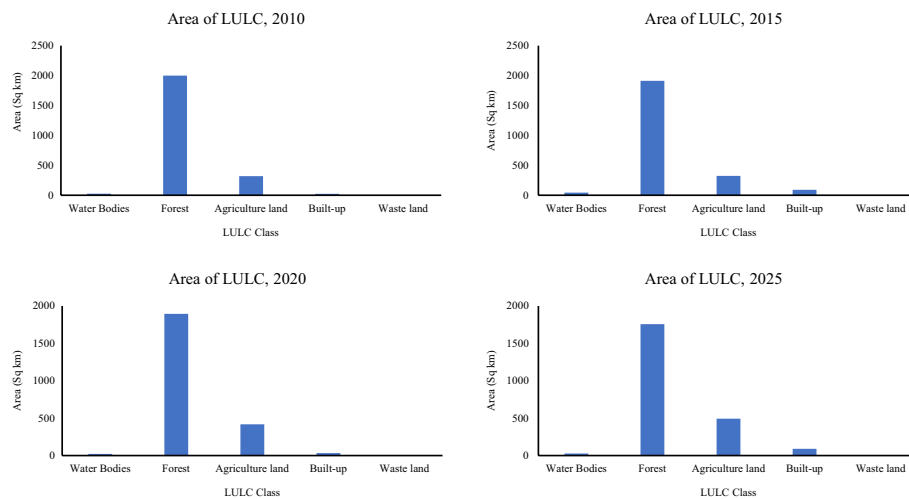
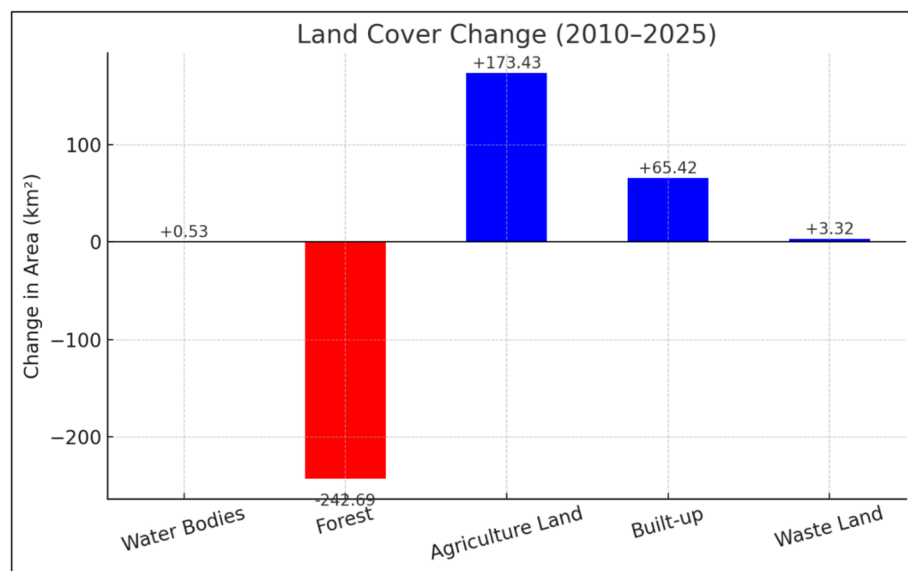
**Fig. 11** Area under different LULC classes (2020–2025)**Fig. 12** Net area changes of land cover classes (2010–2025)

Table 9 Trend statistics for annual mean NDVI and NDWI (2010–2025) over the study area (OLS regression and Mann–Kendall test)

Index	OLS slope/year	OLS (p)	OLS (R ²)	Sen's slope/year	Mann–Kendall (τ)	Mann–Kendall <i>p</i> value
NDVI	0.0070	0.0014	0.5284	0.0064	0.4666	0.0133
NDWI	−0.0048	0.0044	0.4514	−0.0043	−0.4166	0.0274

matrix (Table 8) confirms that the dominant conversion pathway is forest-to-agricultural land, with additional contributions from forest-to-built-up and agricultural land-to-built-up transitions.

4.6 Classification accuracy summary

Overall classification performance for each year indicates robust agreement, with overall accuracy and Kappa statistics summarized in Table 6. Detailed confusion matrices and class-wise accuracy diagnostics for all four years are provided in the Supplementary Material (Tables S1–S4, S10). These diagnostics are used in the Discussion to interpret trends conservatively for classes known to exhibit higher spectral confusion in medium-resolution imagery. Class-wise metrics (Table S10) show consistently high performance for forest and water bodies across years (e.g., 2025 F1 = 0.971–0.977), whereas built-up exhibited lower recall in 2010 (PA = 0.267; F1 = 0.348) due to confusion with agriculture and waste land, but improved substantially in later years (2025 F1 = 0.866). Balanced accuracy (mean recall) increased from 0.772 (2010) to 0.922 (2025), supporting overall robustness while highlighting the need for caution for spectrally mixed classes.

5 Discussion

5.1 Interpreting NDVI and NDWI dynamics

The NDVI maps highlight increasing spatial heterogeneity in vegetation condition across the study area, with localized declines and fragmentation around settlement corridors and conversion hotspots. However, the annual mean NDVI time series shows a statistically significant increasing trend (Fig. 7, Table 9), suggesting that greening in some parts of the landscape (e.g., agricultural expansion, plantations, or recovery in remaining forest/agroforestry mosaics) offsets localized losses in the district-wide mean. NDWI exhibits both marked inter-annual variability in benchmark years (Fig. 5) and an overall significant decreasing trend in annual mean NDWI (Fig. 7, Table 9), consistent with a declining surface-wetness signal. Rainfall patterns derived from CHIRPS v3 (Figs. 6, 8) provide climatic context for this variability and help interpret NDWI fluctuations alongside land-cover change.

5.2 LULC transitions and plausible drivers

The dominant LULC trajectory in Ri-Bhoi district is forest decline coupled with agricultural expansion and a net increase in built-up area. As a mountain–plain transition zone with expanding transport connectivity and mixed land-use mosaics, the district is likely experiencing progressive conversion of forest and fallow patches to cropland, settlements, and associated infrastructure. The non-linear pattern in built-up estimates across intermediate years also suggests that settlement expansion may be spatially clustered and sensitive to class separability in mixed pixels, requiring cautious interpretation.

5.3 Uncertainty and class-wise confusion

Class-wise accuracy diagnostics (Supplementary Tables S1–S4) indicate that some classes, particularly built-up can exhibit higher confusion with spectrally similar surfaces such as bare land and harvested/agricultural patches in medium-resolution optical imagery. Therefore, built-up dynamics are interpreted cautiously and primarily in a relative sense rather than as precise absolute estimates. Forest and agriculture trends are comparatively more stable and consistent with spatial patterns in NDVI and mapped class trajectories, strengthening confidence in the dominant transitions reported.

5.4 Implications for monitoring and planning

The combined LULC and index-based results demonstrate increasing anthropogenic pressure on forested landscapes in Ri-Bhoi district and highlight the value of cloud-based platforms for transparent, reproducible regional monitoring. The workflow and accompanying supplementary diagnostics (confusion matrices and code links) provide a replicable template for similar mountain–plain transition zones where heterogeneous land-cover mosaics and cloud constraints complicate long-term analysis. These outputs can support baseline landscape planning, prioritization of forest conservation/management zones, and improved monitoring of surface-water variability in a monsoon-driven setting. These planning implications align with recent applications that link land transformation, vegetation change, and water-resource planning for decision support [14, 39, 40].

6 Limitations and future research

This study has several limitations. First, inter-annual comparisons involve multiple sensors (Landsat and Sentinel-2) with different spatial resolutions and spectral response functions; despite consistent preprocessing and cloud filtering, these differences may influence class separability and trend comparability. Recent reviews of GEE-based Earth observation and AI workflows emphasize that cross-sensor harmonization (spatial and radiometric) remains a key challenge for multi-temporal analyses and recommend use of harmonized products or cross-sensor adjustment where feasible [11]. Although Sentinel-2 inputs were resampled to 30 m and classification was restricted to common Blue/Green/Red/NIR bands, residual cross-sensor differences—especially in the red and near-infrared bands—may introduce small systematic offsets in indices; therefore, trends are interpreted mainly in terms of direction and relative magnitude rather than precise absolute values. Second, although overall accuracy and agreement metrics are robust (Table 6), class-wise diagnostics show relatively higher confusion for the built-up class in some years (Supplementary Tables S1–S4, S10), and built-up trends should therefore be interpreted conservatively. Third, because the training–validation split is performed at the point level, spatial autocorrelation may slightly inflate accuracy estimates; future work could apply spatially blocked cross-validation and minimum-distance sampling to further strengthen independence between training and validation samples. Fourth, LULC change analysis is based on benchmark-year classifications (2010, 2015, 2020, and 2025) rather than continuous annual LULC mapping; consequently, short-term change events may not be fully captured. Fifth, we did not compute landscape-ecology metrics (e.g., patch density, fragmentation, or edge-density indices) from the classified maps; such metrics could complement class-area change by quantifying spatial configuration

and will be explored in future work. Finally, while rainfall context is provided using CHIRPS v3 benchmark-year maps (Figs. 6, 8) [12], additional hydro-climatic drivers (e.g., drought indices) were not explicitly modelled, and inter-annual NDVI/NDWI variability may reflect both land-cover change and year-specific hydro-meteorological conditions. Future research should consider annual time-series change detection, harmonized multi-sensor calibration, and integration of additional datasets (e.g., SAR for built-up/wetness discrimination, topographic variables, and hydro-climatic drivers) to improve attribution of observed changes and reduce uncertainty in spectrally overlapping classes.

7 Conclusion

This study presents a reproducible, cloud-based framework for assessing multi-temporal land-use/land-cover (LULC) dynamics and associated biophysical trends in a mountain–plain transition zone of northeast India. Using Google Earth Engine and a consistent Random Forest classification approach, LULC distribution and change were quantified for four benchmark years (2010, 2015, 2020, and 2025), complemented by NDVI and NDWI analyses to support interpretation of vegetation and surface-water dynamics.

The results reveal a dominant trajectory of forest decline and agricultural expansion over the 2010–2025 period, accompanied by a net increase in built-up area and pronounced inter-annual variability in surface-water/moisture conditions. NDVI trends indicate increasing spatial heterogeneity and gradual degradation or fragmentation of vegetated landscapes, while NDWI patterns highlight strong year-to-year variability rather than a monotonic trend. Together, these findings reflect increasing anthropogenic pressure on forested landscapes within the Ri-Bhoi district and underscore the sensitivity of transition zones to land-use change.

Methodologically, the study emphasizes transparency and reproducibility by explicitly documenting satellite data selection, preprocessing criteria, classification procedures, and validation diagnostics. Overall accuracy and agreement statistics indicate robust classification performance, while class-wise confusion matrices provided in the Supplementary Material support cautious interpretation of classes prone to spectral overlap, particularly built-up areas. By separating Results, Discussion, and Limitations, the revised manuscript ensures clear distinction between observed patterns, interpretation, and uncertainty.

The proposed workflow demonstrates the utility of cloud-based geospatial platforms for regional environmental monitoring in heterogeneous and data-intensive landscapes. The integration of LULC mapping with NDVI and NDWI trend analysis provides a scalable baseline for future monitoring, planning, and comparative studies in similar mountain–plain transition settings. With further refinement through annual time-series analysis and multi-sensor integration, such approaches can contribute to improved landscape management and evidence-based decision-making in rapidly changing environments.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44288-026-00516-3>.

Supplementary file 1.

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Author contributions

Dr. Subhajit Sarkar (Corresponding Author): Conceptualization, Methodology, Data Collection, Supervision, and Writing—Original Draft. Longkiri Terang: Software Analysis, Data Processing, and Writing. Both authors approved the final manuscript.

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Data availability

All datasets used in this study are publicly available through the Google Earth Engine (GEE) platform (<https://earthengine.google.com/>). The analysis was carried out entirely within the GEE environment using open-access Landsat and Sentinel surface reflectance products. The complete GEE scripts used for preprocessing, compositing, Random Forest classification, accuracy assessment, and trend analyses are provided in the Supplementary Material (Section S3).

Declarations

Ethical approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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